MUCHUKUNTLA BRAHMANAIDU

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Professional Summary

Data Scientist specializing in machine learning, computer vision, and NLP, building end-to-end predictive systems and scalable ML pipelines. Skilled in Python and SQL, with experience in model evaluation, feature engineering, and real-time deployment through internship and project work.

Education

Annamacharya Institute Of Technology & Sciences (JNTUA) — **Dec 2021 – Apr 2025**
Bachelor of Technology in Computer Science — CGPA: 7.9 / 10

TECHNICAL SKILLS

- **Languages & Querying:** Python, SQL (PostgreSQL)
- **Machine Learning & DL:** Scikit-learn, TensorFlow, PyTorch, Model Evaluation, Feature Engineering
- **Data Science Stack:** Pandas, NumPy, SciPy, Matplotlib, Seaborn
- **Mathematics & Stats:** Descriptive Statistics, Hypothesis Testing, A/B Testing
- **Visualization & BI:** Power BI, Excel, Data Storytelling

Experience

Machine Learning Engineer Intern | *Zithara.AI (Propel5000)* — **Jan 2025 – Apr 2025**

- **Visual Search System:** Developed a search-by-image feature for an e-commerce platform, enabling similarity-based retrieval across a 5,000+ SKU jewelry catalog.
- **Computer Vision Modeling:** Built a PyTorch-based ResNet50 embedding pipeline to extract discriminative visual features capturing fine-grained gold textures and designs.
- **Scalable Inference:** Integrated FAISS for high-speed similarity search (<200ms latency) and deployed the ML pipeline as a Python microservice for real-time application use.
- **Model Evaluation & Tuning:** Assessed retrieval quality using similarity-based evaluation (e.g., top-k results) and iteratively refined preprocessing and embedding consistency to improve result relevance.

Projects

Market Pulse – Customer Segmentation & Revenue Analysis System | *Python, Pandas, Scikit-learn*

- Built an end-to-end data analysis and clustering pipeline to segment customers using transactional and behavioral features, including feature scaling and RFM-based feature engineering.
- Applied unsupervised learning techniques (KMeans, Hierarchical Clustering) and evaluated cluster quality using silhouette score to identify distinct customer groups.
- Analyzed revenue contribution and behavioral patterns across segments to generate data-driven marketing and retention strategy recommendations.

Smart Hire AI – Resume Screening & Skill Matching System | *Python, NLP, Sentence Transformers*

- Developed an AI-powered resume screening system that matches candidate resumes with job descriptions using text preprocessing and embedding-based similarity scoring.
- Generated semantic embeddings using transformer-based models and computed cosine similarity to rank candidates based on job relevance.
- Designed a skill gap analysis module to identify missing competencies and provide structured insights to support data-driven hiring decisions.

Vision Guard – AI-Based Real-Time Defect Detection System | *Python, PyTorch, OpenCV*

- Engineered an end-to-end computer vision pipeline to detect product defects from image datasets using preprocessing, augmentation, and feature extraction techniques.
- Trained and evaluated deep learning models (CNN/ResNet) using accuracy, precision, and recall metrics to classify defective and non-defective samples.
- Optimized inference performance and deployed the trained model as a lightweight API for real-time quality inspection applications.

CERTIFICATIONS

- **Google Data Analytics Professional Certificate** – Coursera (Tools: SQL, Tableau, R)
- **Data Science & Machine Learning Bootcamp** – Udemy